# Secure User Authentication in Web Applications

## Using HTML, CSS, JavaScript, PHP, MySQL & Sessions

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## 1. Introduction

User authentication verifies the identity of a user before giving access to a web application.
It helps to protect user data and prevents unauthorized access.

## 2. Problem

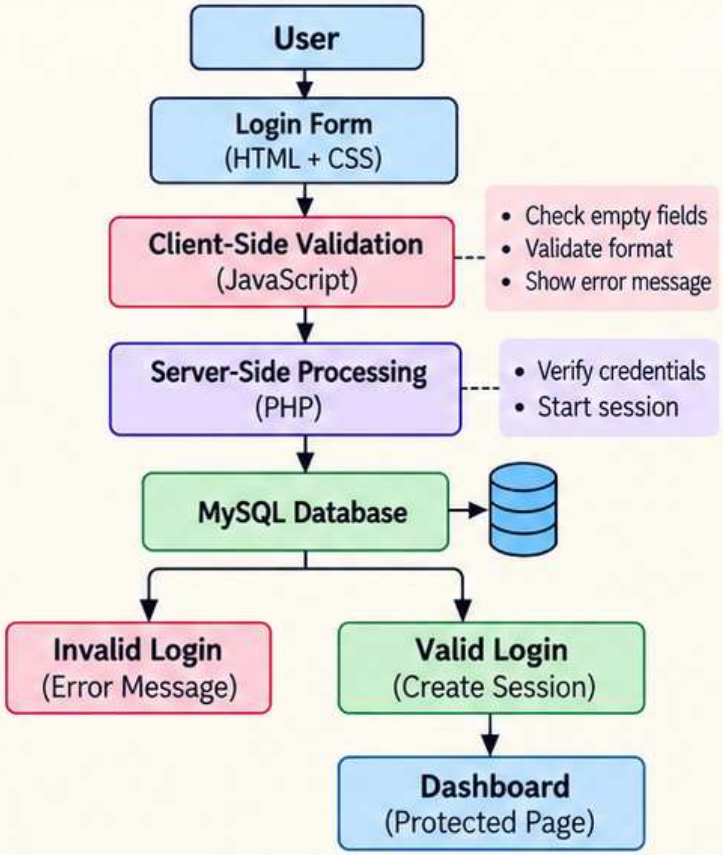
- Unauthorized access to web pages
- Invalid or incorrect login details
- Security threats to user data
- Need to maintain user session



## 3. Technologies Used

| HTML | CSS | JavaScript | PHP | MySQL |
|---|---|---|---|---|
| Structure of web pages | Design and layout | Client-side validation | Server-side processing | Data storage |

## 4. System Architecture / Workflow




## 5. Login Form (UI)




## 6. Output Screens




## 7. Database Table (users)

| id | username | password | email |
|---|---|---|---|
| 1 | vaidehi | ******** | vaidehi@gmail.com |
| 2 | meet | ******** | meet@gmail.com |
| 3 | riya | ******** | riya@gmail.com |

## 8. Code Snippets

### HTML (Login Form)

```
<form method="post" action="login.php">
  <input type="text" name="username"
    placeholder="Enter username" required>
  <input type="password" name="password"
    placeholder="Enter password" required>
  <button type="submit">Login</button>
</form>
```

### JavaScript (Validation)

```
function validate() {
  let user = document.
      getElementById("username").value;
  let pass = document.
      getElementById("password").value;
  if(user=="" || pass==""){
    alert("Please enter username
     and password!");
     return false;
  }
}
```

### PHP (Login & Session)

```
<?php
sesion_start();
$conn = new mysqli("localhost","root","","students");
$user = $_POST['username'];
$pass = $_POST['password'];
$sql = "SELECT * FROM users
        WHERE username='$user'
        AND password='$pass'";
$result = $conn->query($sql);
if($result->num_rows > 0){
   $_SESSION["user"] = $user;
   header("Location: dashboard.php");
} else {
   echo "Invalid username or password!";
?>
```

## 9. Security Practices

- Validate input on both client-side and server-side
- Use password hashing (e.g., password_hash())
- Use pessions to maintain login state
- Use prepared statements to prevent SQL injection
- Show proper error messages
- Logout and destroy session when user exits

## 10. Conclusion

The system provides a simple and secure way to authenticate users using HTML, CSS, JavaScript, PHP, MySQL and sessions. It helps to protect user data and restrict unauthorized access.